

Schools in England and Wales, 2020-25

Reasons for choosing the datasets

I have chosen the data on schools' basic information and inspection results to find out the general picture of schools in England and Wales:

- How many students do schools have?
- What are the geographical differences?
- How well are schools doing?
- Are there links between inspection success and other school features?

Since the first dataset only includes one continuous numerical variable, I have included a population dataset, which will enable me to ask further questions:

- Are inspection results linked to size of local population?
- Is school size linked to size of local population?

I also wanted to look not just at how many people in the local area, but the population density, so I also included a third data set which addresses that.

Dataset 1: Schools data Feb 2025

Overview

- Basic information about all schools
- Summary results of the latest inspection (and a link to the report)

Source

[State-funded school inspections and outcomes: management information - GOV.UK](#)

This data is published by Ofsted, the government's inspectorate for schools. Since it is government data, it can be assumed to be a trustworthy data source.

Collection

Ofsted collects basic information about the schools from the schools themselves. Schools are likely to give accurate information and most of it is categorical data, which is less subject to input errors. The inspection data is collected or decided by Ofsted. There is a small chance of input errors if the data is entered manually, which seems likely.

Contents

Basic information about schools includes:

- Ofsted Phase (Primary, Secondary etc)
- Local authority
- Total number of pupils

Summary results of the latest inspection include:

- Inspection start date
- Overall effectiveness (graded 1 to 4 or “not judged”)
- Quality of education (graded 1 to 4 or 9)

There is also similar data about the previous inspection

Timeliness

The dataset was published on 28th February 2025, so the data is up to date. However, schools are not all inspected at the same time, and the length of time between inspections can vary depending on the school’s performance and previous inspection outcomes. Therefore the latest inspection results could be as recent as one month ago or as long as seven years ago, with most schools being inspected every 2½ to 4 years.

Dataset 2: Population data

Overview

Population figures for each local authority

Source

[Estimates of the population for England and Wales - Office for National Statistics](#)

The data comes from the Office for National Statistics. The Office for National Statistics is the executive office of the UK Statistics Authority, a non-ministerial department which reports directly to the UK Parliament. As such, it can be assumed to be a trustworthy data source.

Collection

The numbers are not actual population figures, but estimates, based on census data from 2021 and predictive modelling.

Contents

Population figures for each local authority, subdivided by single year of age and sex

Timeliness

The dataset was published on 15th July 2024, using data from the 2021 census, the most recent census.

Dataset 3: Population density

Contents

Population density for each local authority

Source

[Population density - Office for National Statistics](#)

The data comes from the Office for National Statistics. As stated above, it can be assumed to be a trustworthy data source.

Collection

These are actual figures, taken from census data and geographic data. There is a small chance of input error.

Timeliness

The dataset was last updated on 16th February 2023, and it uses 2021 census data.

Data Profile

Dataset 1: Schools_2025

Column Heading	Description/Cleaning steps taken
Web link (opens in new window)	<i>deleted – not useful for analysis</i>
URN	a unique reference number for each school
LAESTAB	<i>deleted – not useful for analysis</i>
School name	<i>deleted – for privacy reasons</i>
Ofsted phase	Value counts for Ofsted phase: Ofsted phase Primary 16719 Secondary 3412 Special 1090 Nursery 379 PRU 301 Name: count, dtype: int64
Type of education	<i>deleted – not useful for analysis</i>
School open date	

Admissions policy	Value counts for Admissions policy: Admissions policy Not applicable 17612 Non-selective 3418 Unknown 707 Selective 164 Name: count, dtype: int64
Sixth form	Value counts for Sixth form: Sixth form Does not have a sixth form 17358 Has a sixth form 2665 Not applicable 1872 Not recorded 6 Name: count, dtype: int64
Designated religious character	<i>deleted – not useful for analysis</i>
Religious ethos	<i>deleted – not useful for analysis</i>
Faith grouping	<i>deleted – not useful for analysis</i>
Ofsted region	<i>deleted – not useful for analysis</i>
Region	Value counts for Region: Region South East 3369 North West 3182 London 2557 East of England 2548 West Midlands 2420 South West 2367 Yorkshire and the Humber 2237 East Midlands 2079 North East 1142 Name: count, dtype: int64
Local authority	
Parliamentary constituency	<i>deleted – not useful for analysis</i>
Multi-academy trust UID	<i>deleted – not useful for analysis</i>
Multi-academy trust name	<i>deleted – not useful for analysis</i>
Academy sponsor UID	<i>deleted – not useful for analysis</i>
Academy sponsor name	<i>deleted – not useful for analysis</i>
Postcode	
The income deprivation affecting children index (IDACI) quintile	89 missing entries – records deleted ordinal data - takes values 1-5 Value counts for The income deprivation affecting children index (IDACI) quintile: The income deprivation affecting children index (IDACI) quintile 4.0 4472 5.0 4468 3.0 4418 2.0 4328 1.0 4215 Name: count, dtype: int64
Total number of pupils	min 0 was removed when the 89 missing entries were

	integers but counts as continuous data										
Statutory lowest age	<i>deleted – not useful for analysis</i>										
Statutory highest age	<i>deleted – not useful for analysis</i>										
Latest ungraded inspection number since last graded inspection	<i>deleted – not useful for analysis</i>										
Does the ungraded inspection relate to the URN of the current school?	<i>deleted – not useful for analysis</i>										
URN at time of the ungraded inspection	<i>deleted – not useful for analysis</i>										
LAESTAB at time of the ungraded inspection	<i>deleted – not useful for analysis</i>										
School name at time of the ungraded inspection	<i>deleted – for privacy reasons</i>										
School type at time of the ungraded inspection	<i>deleted – not useful for analysis</i>										
Number of ungraded inspections since the last graded inspection	<i>deleted – not useful for analysis</i>										
Date of latest ungraded inspection	<i>deleted – not useful for analysis</i>										
Ungraded inspection publication date	<i>deleted – not useful for analysis</i>										
Did the latest ungraded inspection convert to a graded inspection?	<i>deleted – not useful for analysis</i>										
Ungraded inspection overall outcome	<i>deleted – not useful for analysis</i>										
Number of monitoring and urgent inspections since last graded inspection	<i>deleted – not useful for analysis</i>										
Inspection number of latest graded inspection											
Inspection type											
Inspection type grouping											
Event type grouping	<i>deleted – not useful for analysis</i>										
Inspection start date											
Publication date											
Does the latest graded inspection relate to the URN of the current school?	<i>deleted – not useful for analysis</i>										
URN at time of latest graded inspection	<i>deleted – not useful for analysis</i>										
LAESTAB at time of latest graded inspection	<i>deleted – not useful for analysis</i>										
School name at time of latest graded inspection	<i>deleted – for privacy reasons</i>										
School type at time of latest graded inspection											
	Overall effectiveness <table> <tr> <td>2</td><td>16053</td></tr> <tr> <td>1</td><td>2590</td></tr> <tr> <td>3</td><td>1552</td></tr> <tr> <td>Not judged</td><td>1378</td></tr> <tr> <td>4</td><td>270</td></tr> </table> Name: count, dtype: int64	2	16053	1	2590	3	1552	Not judged	1378	4	270
2	16053										
1	2590										
3	1552										
Not judged	1378										
4	270										
Overall effectiveness											
Category of concern	<i>deleted – not useful for analysis</i>										

Quality of education	Quality of education 9.0 9978 2.0 8500 3.0 1750 1.0 1380 4.0 194 NaN 99 Name: count, dtype: int64
Behaviour and attitudes	Behaviour and attitudes 9.0 9978 2.0 8277 1.0 2616 3.0 788 4.0 143 NaN 99 Name: count, dtype: int64
Personal development	Personal development 9.0 9978 2.0 8149 1.0 3028 3.0 579 NaN 99 4.0 68 Name: count, dtype: int64
Effectiveness of leadership and management	Effectiveness of leadership and management 2.0 16510 1.0 3650 3.0 1382 4.0 260 NaN 99 Name: count, dtype: int64
Safeguarding is effective?	<i>deleted – not useful for analysis</i>
Early years provision (where applicable)	<i>deleted – not useful for analysis</i>
Sixth form provision (where applicable)	<i>deleted – not useful for analysis</i>
Previous graded inspection number	
Previous inspection start date	
Previous publication date	
Does the previous graded inspection relate to the URN of the current school?	<i>deleted – not useful for analysis</i>
URN at time of previous graded inspection	<i>deleted – not useful for analysis</i>
LAESTAB at time of previous graded inspection	<i>deleted – not useful for analysis</i>
School name at time of previous graded inspection	<i>deleted – for privacy reasons</i>
School type at time of previous graded inspection	

Previous graded inspection overall effectiveness	Previous graded inspection overall effectiveness 2.0 9670 3.0 6390 1.0 3905 4.0 1231 NaN 705 Name: count, dtype: int64
Previous category of concern	<i>deleted – not useful for analysis</i>
Previous quality of education	Previous quality of education 9.0 19930 3.0 1020 NaN 705 4.0 131 2.0 113 1.0 2 Name: count, dtype: int64
Previous behaviour and attitudes	Value counts for Previous behaviour and attitudes: Previous behaviour and attitudes 9.0 19930 2.0 712 NaN 705 3.0 440 4.0 104 1.0 10 Name: count, dtype: int64
Previous personal development	Previous personal development 9.0 19930 2.0 836 NaN 705 3.0 354 4.0 51 1.0 25 Name: count, dtype: int64
Previous effectiveness of leadership and management	Previous effectiveness of leadership and management 2.0 10475 3.0 5270 1.0 4411 4.0 1040 NaN 705 Name: count, dtype: int64
Previous safeguarding is effective?	<i>deleted – not useful for analysis</i>
Previous early years provision (where applicable)	<i>deleted – not useful for analysis</i>
Previous sixth form provision (where applicable)	<i>deleted – not useful for analysis</i>

Note: Ofsted use a 4-point grading scale in all inspections to make the principal judgements:

- grade 1 – outstanding
- grade 2 – good
- grade 3 – requires improvement
- grade 4 – inadequate

Dataset 2: Population data

Contains columns for local authorities and total population, divided by age (from 0 to 90+ individually) and by sex. Population figures were object types and had to be redefined as integers, first removing the commas, in order to aggregate figures into numbers of children at each school level.

Dataset 3: Densities

Contains columns for local authorities and a figure giving population per square kilometre.

Limitations and ethics

Population figures are estimates based on the recent census, as mentioned above.

School names were removed for privacy reasons.

It is possible that some schools might still be identifiable by local authority and school size.

No information about staff numbers or financial data for schools was available.

Questions to explore

Is size of school related to location?

Is inspection result related to size of school? location? phase?

Are inspection results changing over time? Are more recent results better than previous, on the whole?

What can be said about schools which have improved their inspection grading? Any similarities in terms of size, location or phase?

How are the different phases of education doing in terms of inspection grading?

Do schools do better at one particular type of inspection outcome?

Are children and schools evenly distributed about the country or are there areas where the density of school children is higher? Is this true for both primary and secondary age children?

Does school size depend on population density in the local authority?